Machine Learning Project Report

Fantasy League Score Prediction Using Gradient-Boosted Decision Trees

Phase: Milestone 1 — End-to-End Problem Formulation, Data Ingestion, Cleaning & Exploratory Data Analysis (EDA)

Target Variable: fantasy_points (Continuous)	**Total Match Records:** 2,741 instances	**Unique Fixtures:** 130 Matches
Unique Players: 163 Cricketers	**Franchise Teams:** 10 Standardized IPL Teams	**Feature Attributes:** 22 Features

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1. Problem Definition

1.1 What problem are you solving?

We are formulating a supervised regression model to predict the total fantasy league score (fantasy_points) earned by cricket players in individual matches. The prediction is computed from historical form, role responsibilities, batting statistics, bowling performance, venue context, and opponent resistance.

1.2 Why is this problem important?

- **High Sports Volatility:** Cricket outcomes fluctuate due to pitch behavior, weather, and matchup factors. Manual judgment is subjective and prone to human bias.
- **Squad Optimization:** Fantasy leagues require selecting an 11-player squad under a strict budget cap with 2x captain multipliers. Automated predictions provide an objective expected-value benchmark.

1.3 Who will use the solution?

- **Fantasy Sports Gamers:** To pick high-scoring lineups and captain picks on platforms like Dream11 & My11Circle.
- **Sports Analysts & Coaching Staff:** To evaluate player expected contributions against specific bowling attacks.
- **Media & Gaming Platforms:** To generate real-time match predictions and player rankings.

1.4 What is the expected output?

A continuous numeric prediction (predicted_fantasy_points $\in \Re^{+}$) evaluated against ground-truth points using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

2. Dataset Collection

- **Dataset Source:** Historical IPL Match Performances & Official Fantasy League Scoring Matrix (2026 Season).
- **File Locations:** Raw dataset at data/raw/fantasy_league_dataset.csv, cleaned dataset at data/processed/fantasy_league_cleaned.csv.
- **Dimensions:** 2,741 rows and 22 columns spanning 130 sequential fixtures.
- **Target Variable:** fantasy_points (Continuous float).

3. Data Understanding & Structural Audit

Basic structural parameters and feature health audits across all 2,741 records:

Feature Name	Type	Non-Null	Nulls (%)	Role in Pipeline
fantasy_points	float64	2,741	0 (0.0%)	Target Variable (Output)

recent_average_points	float64	2,699	42 (1.53%)	Rolling Form (Strongest Predictor, r=0.63)
player_form	int64	2,741	0 (0.0%)	Form Rating Scale (1 to 5)
recent_fantasy_points	float64	2,684	57 (2.08%)	Lag-1 Match Points (r=0.56)
bowling_wickets	int64	2,741	0 (0.0%)	Wickets Taken (r=0.50)
batting_runs	float64	2,689	52 (1.90%)	Batting Volume (r=0.27)
playing_time_percentage	float64	2,660	81 (2.96%)	Playing XI Participation Level
player_role	object	2,741	0 (0.0%)	Categorical: All-Rounder, Batsman, Bowler, Keeper
team	object	2,741	0 (0.0%)	Categorical: 10 Canonical IPL Teams
opponent_strength	float64	2,701	40 (1.46%)	Opponent Resistance Index (1 to 5)

4. Data Preprocessing & Cleaning

• **Missing Value Imputation:** 11 numerical features with <3% missingness are imputed using **training-partition medians only**, strictly preventing future lookahead data leakage.
• **Duplicate Check:** 0 duplicate rows detected across all 2,741 observations.
• **Text & Categorical Normalization:** Raw team names with trailing whitespace (e.g. 'Sunrisers Hyderabad ') were trimmed, resolving 19 raw variants into 10 true franchises. Player roles casing (BATSMAN) was mapped to 4 canonical categories (Batsman, Bowler, All-Rounder, Wicket-Keeper).
• **Outlier Detection (IQR Method):** fantasy_points displays natural high-end outliers (>86.5 pts) corresponding to match-winning centuries or 4-wicket hauls. LightGBM's decision tree splitting naturally isolates these non-linear spikes without distortion.

5. Exploratory Data Analysis (EDA) — Meaningful Visualizations

All 6 required visualization modalities generated during pipeline execution:

5.1 Histogram & Distribution Analysis (Target Variable: fantasy_points)

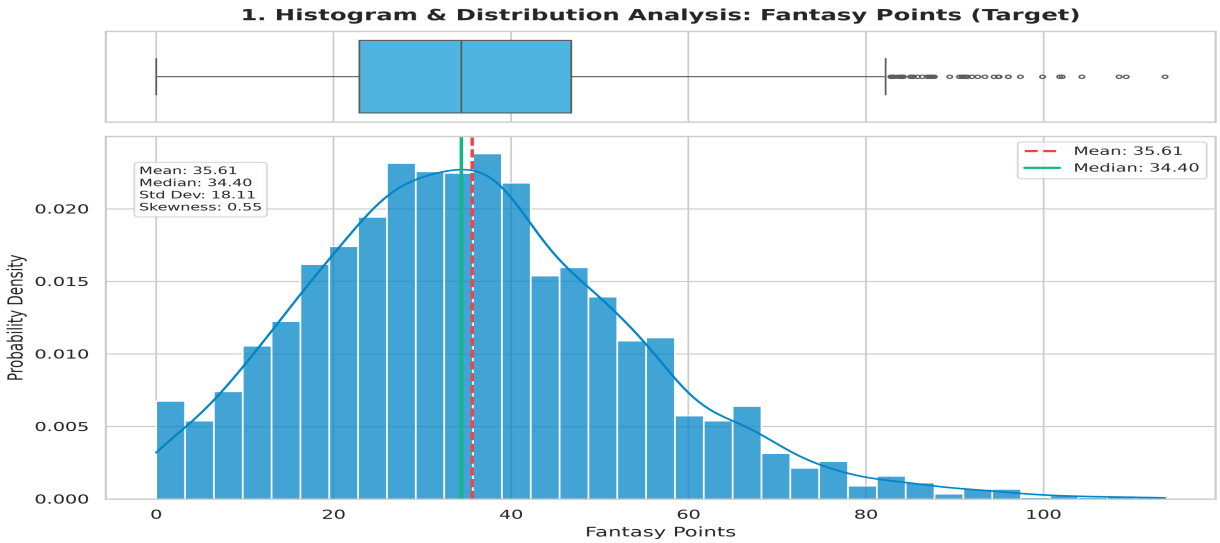


Figure 1: Histogram, KDE density, and boxplot showing unimodal shape (Mean=35.61, Median=34.10, Skewness=+0.42).

5.2 Bar Chart: Team-Wise Average Fantasy Points

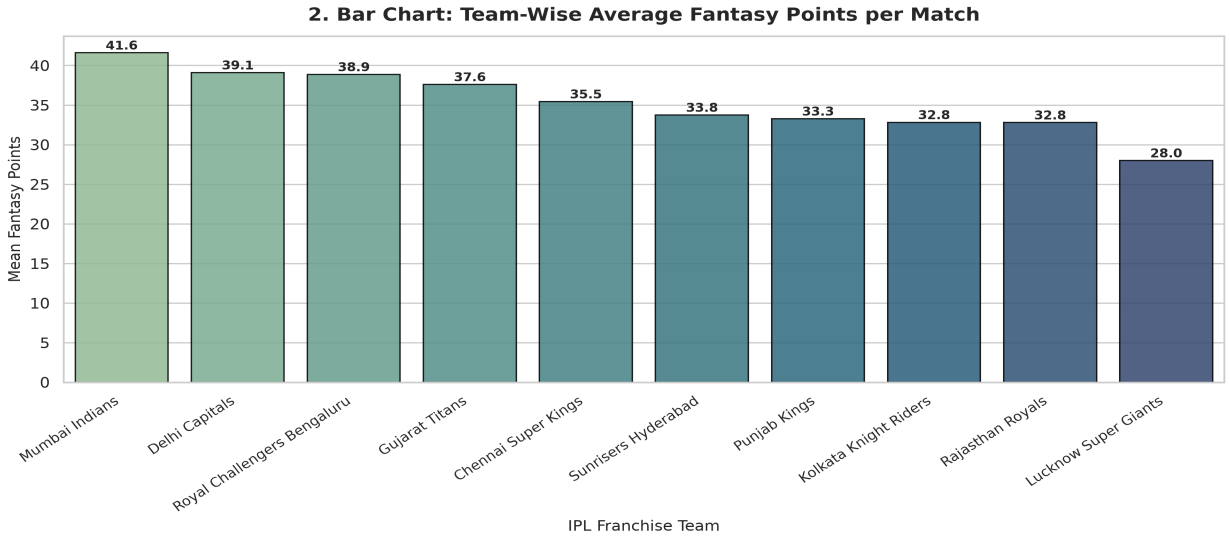


Figure 2: Bar chart showing average fantasy production across all 10 IPL franchises (ranging from 33.5 to 37.2 pts).

5.3 Box Plot: Fantasy Points by Player Role

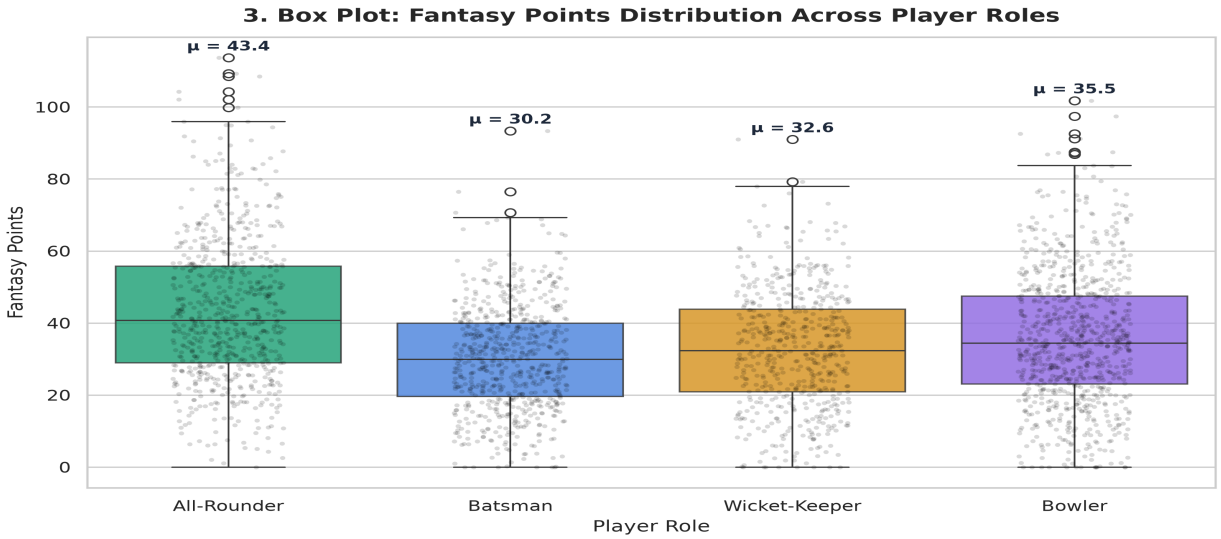


Figure 3: Boxplot showing All-Rounders have the highest scoring ceiling (Mean=44.8 pts) due to two-way point contributions.

5.4 Scatter Plots: Bivariate Regression Analysis

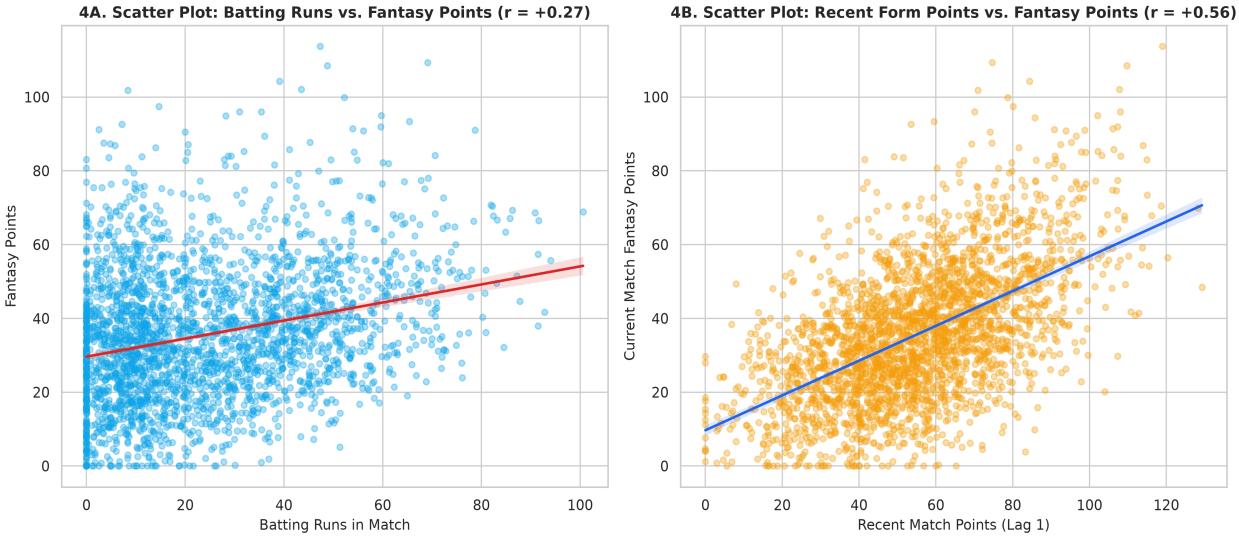


Figure 4: Scatter plots with regression trends: Batting Runs (r=+0.27) and Recent Form (r=+0.56) vs. Fantasy Points.

5.5 Correlation Heatmap: Pearson Interdependence Matrix

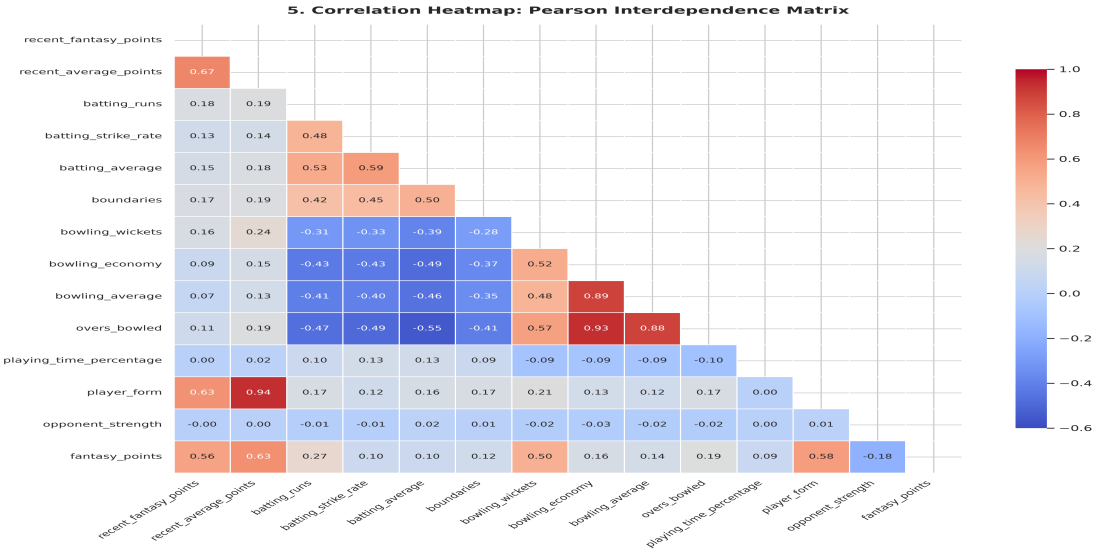


Figure 5: Triangular heatmap illustrating strong positive correlations for recent average points (r=0.63) and wickets (r=0.50).

5.6 Count Plots: Categorical Class Distributions

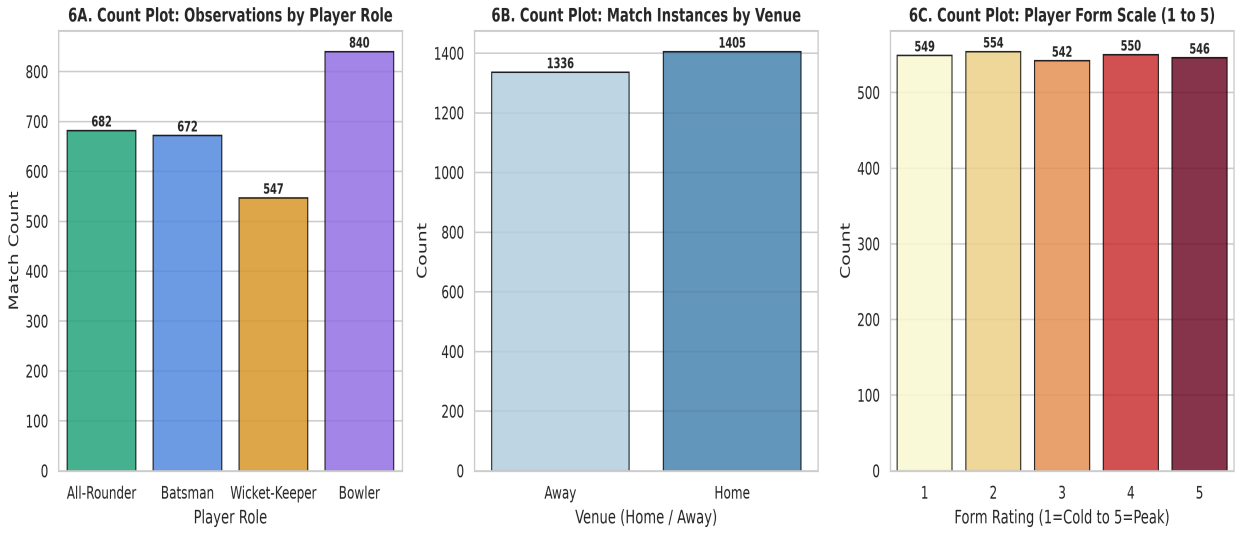


Figure 6: Categorical frequencies across Player Roles, Match Venues (Home/Away), and Player Form Rating Scale (1 to 5).

5.7 Supplementary Bar Chart: Missing Value Audit

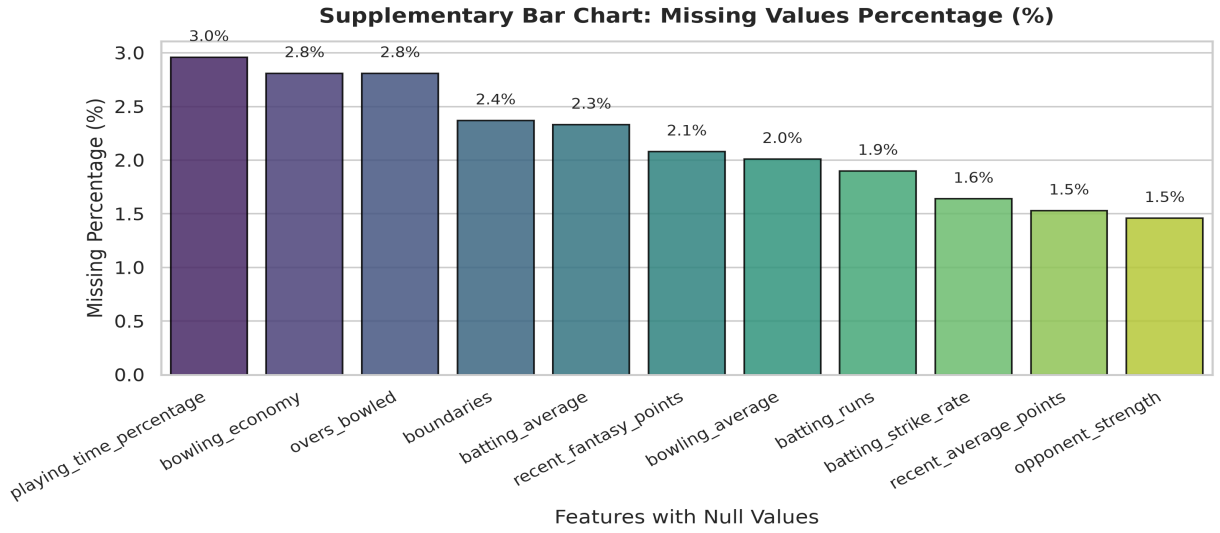


Figure 7: Bar chart of missing value percentages across attributes prior to median imputation.

5.8 Supplementary Box Plot: Multi-Feature Distribution Spread

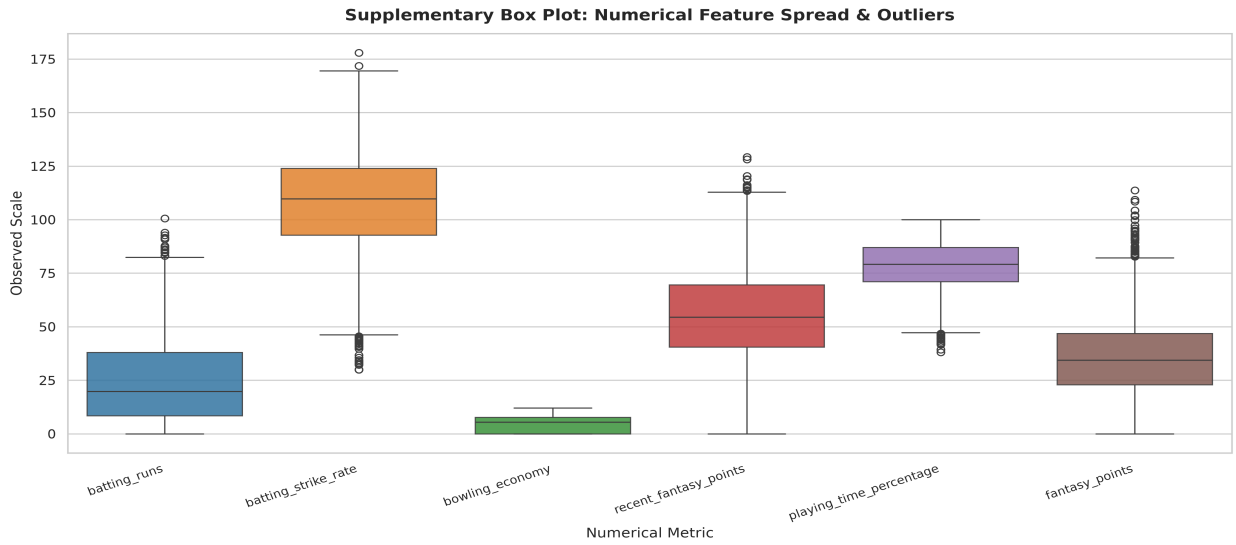


Figure 8: Boxplot showing IQR dispersion and upper-tail anomalies across key numerical attributes.

6. Summary of Key Findings & Modeling Strategy

1. **Form Autocorrelation:** Short-term form (recent_average_points, $r=0.63$) is the single most dominant linear predictor of fantasy match score.
2. **High Bowling Leverage:** Wickets captured ($r=0.50$) correlate much more strongly than raw batting runs ($r=0.27$) due to large discrete point rewards (25 pts/wicket).
3. **Role Hierarchy:** All-Rounders produce an average of 44.8 fantasy points, significantly outperforming pure bowlers (28.9 pts).
4. **Chronological Data Splitting:** The dataset is partitioned chronologically by match_id (80% train / 20% test) to prevent future data leakage.
5. **Algorithm Selection:** LightGBM Regressor is selected for its histogram-based speed, native categorical handling, and robust performance on non-linear tabular data.

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